Validation report — Nhill 3.0.0

Français : ../fr/rapport-validation.md

Version 3.0.0 — plans, documents, data, deployment

Date	29 October 2026
Version	3.0.0 (`project(nhill VERSION 3.0.0)`), branch `release/3.0.0`, not published
Validated code commit	`18d7054` (later commits only change documentation)

Tests run (continuous integration of commit `18d7054`)

Suite	Where	Result	Evidence
`common`	CI Ubuntu 24.04	**22/22**	run 40826508382, `ci-results/ci`
`server` (plans and rights for every plan and every role, limits, computed access, documents, till, erasure of a business, retention, searches, forgotten password…)	CI Ubuntu 24.04 (PostgreSQL 16); workspace	**121 passed, 1 skipped** (`Samples.GenerateDocumentSamples`) out of 122	same
`setup`, `client`	CI	**6/6**, **15/15**	same
`keychain` — real Secret Service	CI Ubuntu 24.04	**3/3** (+ init/cleanup)	`ci-results/ci`, `logs/keychain-linux.txt`
`keychain` — real Windows Credential Manager	CI `windows-2022`	**3/3**	run 40826508391
`keychain` — Android Keystore	Android 11 emulator (CI)	phases `write`, `read`, `logout`, `forget`: **OK**	run 40826508406, `ci-results/android-keychain`
Ubuntu 24.04 and Debian 13 `.deb` packages, `apt` installation then removal	CI (clean containers)	**passed**	run 40826508393
VPS kit: `provision.sh` on a fresh Ubuntu 24.04 machine, then acceptance	CI	**12/12 checks** (local ports, firewall, secrets, TLS and HSTS through Caddy, monitoring, encrypted backup restored off the server, update, rollback, refusal of a newer database, first administrator)	`ci-results/vps-test`
Windows installers, silent installation	CI `windows-2022`	passed	`ci-results/windows`
APK: third-party notices present	CI	`assets/THIRD-PARTY-NOTICES.txt` present	`ci-results/android`
API and database references up to date	CI	passed	`ci-results/ci`, `ci-results/docs-website`
PDFs up to date, website, forms backend (SQLite, PostgreSQL)	CI	passed; 24 pages, 0 problems	run 40826508436
QML check	workspace, CI	100 files, 0 problems, 0 untranslated texts	`scripts/qa/qml-check.py`
`nhill.online` domain	CI	**expected failure**: Hostinger parking page, no MX, SPF or DMARC	`ci-results/domain-check`

SHA-256 of the deliverables built by CI (`18d7054`):

```
fe68846cf61ff330f5ca8f332916d38685417bba272f4213c14f6d82cc19da9b  windows/nhill-business-3.0.0-setup.exe
9d5d55a841850ee9dc3a8d8911fd84398f0a01b10c472384e1dbb65b899be763  windows/nhill-pos-3.0.0-setup.exe
eb654f3bfa0ef7e39785f03c8bacc8f466e281556744686f8ca6d3da8aa498e6  windows/nhill-console-3.0.0-setup.exe
500d06e310fe7a68d5b29f5a2a5ddab5c6055b54274300eb10f671fde573c0e8  android/nhill-business-3.0.0-arm64-v8a.apk
456ca24392c57fd860b3096b05d564c0e828b1afad97df6c61be2bd2d542757f  linux/ubuntu-24.04/nhill-server_3.0.0-
1_amd64.deb
f9cddb2f63854b65423757d521dd2eea97d4fc3a287df904b8ae23c9d4366c0f  linux/debian-13/nhill-server_3.0.0-1_amd64.deb
cb60e13fa18e09f1430f9d983bb86bdd6fee141c4fe5e48adeae67fa30d34c87  website/nhill-website-3.0.0.tar.gz
```

Full list: `release/V3.0.0/checksums/SHA256SUMS`.

Defects found and fixed during the 3.0 validation

Defect	Fix
Erasure of a business promised by the terms but impossible (documents and log immutable for the application)	migration 0014, controlled erasure function, console screen, tests
`%` and `_` interpreted in searches	migration 0015, `nh_like_contains`
VPS acceptance: inactive PostgreSQL umbrella unit, internal certificate at 0 days, restore test unreadable for `nobody`, backup checksum with absolute path	`pg_isready` check, adapted threshold, script copy, checksum on the file name
POS 1366×768 screenshots missing (Xvfb display not ready)	wait for the X socket before launching
Android Keystore test: “Can’t find service: activity” right after boot	wait for the service and retry `am` commands
APK without QtKeychain’s BSD notice (statically linked)	notices included and checked at build time
Sales file: bracket fields, fictitious tracker rows, outdated “8-week pilot” offer	declared merge fields, empty files, 30-day trial everywhere

Not tested

- Real deployment on the Hostinger VPS (no access); `nhill.online` website not published.
- Real Android phone, user Windows 10/11 machine, e-mails through a real SMTP relay.
- Upgrade of a 2.1 database holding real data to 0012–0015 (tests apply 0001–0015 to an empty database).
- Hiding of features in the interface for each plan (checked server-side by the `Plans.*` tests; in the applications by code reading and static check).

Pending decisions and human actions

- R1 (cpp-httplib of the Ubuntu 24.04 package) and R2 (Qt 6.5.3 on Windows and Android): choice of the fix (03-security-audit.md § 4).
- Website host (legal notice), sender address of automatic e-mails, mail DNS records (MX, SPF, DMARC).
- Signing certificates (Authenticode, Android release key), legal review (points V-01 to V-14), external audit.

Version 2.1.0 — remembered session and forgotten password

Date	27 October 2026
Version	2.1.0 (`project(nhill VERSION 2.1.0)`), branch `release/2.1.0`, unreleased
Validated code commit	`f8524fdd048c7f8462165925aae30d9b4d4357f1` (later commits change only the documentation, the marketing website and the commercial documents)

Tests run

Suite	Where	Result	Evidence
`common`	CI Ubuntu 24.04, workspace	**22/22**	run 40541503872
`server` (including 16 forgotten-password tests: reset by the administrator, e-mailed code, SMTP delivery to a local fake relay, SMTP configuration)	CI Ubuntu 24.04 (PostgreSQL 16); workspace	**99 passed, 1 skipped** (`Samples.GenerateDocumentSamples`) out of 100	same

Suite	Where	Result	Evidence
`setup`, `client`	CI	**6/6**, **13/13**	same
`keychain` — real Secret Service (gnome-keyring, two processes)	CI Ubuntu 24.04	**5/5** (write then read back in a second process, sign-out, nothing stored without “Remember me”)	`ci-results/ci`, `logs/keychain-linux.txt`
`keychain` — real Windows Credential Manager, `NHILL_KEYCHAIN_REQUIRED=1`	CI `windows-2022`	**5/5**	run 40541504066, `keychain-test-log.txt`
QtKeychain linked into each Windows application	CI `windows-2022`	`CredReadW`, `CredWriteW`, `CredDeleteW` imported from ADVAPI32; no QtKeychain DLL	same
Silent install and uninstall (Nhill POS, Nhill Business)	CI `windows-2022`	passed; installed executable identical to the checked one (SHA-256)	same
`keychain` — real Android Keystore	Android 11 x86_64 emulator (CI)	phases `write`, `read` (new launch), `logout`, `forget`: **OK**	run 40541503866, `ci-results/android-keychain`
Nhill Business APK	CI	built; QtKeychain Keystore code present in the library	same
Ubuntu 24.04 and Debian 13 `.deb` packages, clean install with `apt` then removal	CI (`ubuntu:24.04`, `debian:trixie` containers)	**passed**	run 40541504028, `ci-results/deb-*`, `ci-results/install-*`
QML static check	workspace	98 files, 0 problems, 0 untranslated text	`scripts/qa/qml-check.py`
Marketing website	workspace	24 pages, 0 problems; contrast 0 failures (16 pages × 2 widths)	`check_site.py`, accessibility probes
Financial model	workspace	documents up to date	`subscription_model.py --check`

Digests of the Windows installers built by the CI (`f8524fd`):

```
e7b12542145330b7f9f5afc1131bd1ba6ed666607739694625 52a88a37108b28  nhill-business-2.1.0-setup.exe
fd7048f0b4d6ec1031019b21242ea21b16cd6c3a8bded5a53e8a01b3ff2837b0  nhill-console-2.1.0-setup.exe
64e0493dc921090209b6a459c021da9c76f11ec4835159711a442d5c843278d3  nhill-pos-2.1.0-setup.exe
```

Defects found and fixed during the 2.1 validation

Defect	Fix
`windeployqt` treated `qt6keychain.dll` as a Qt module and looked for it in Qt’s folder	QtKeychain linked statically on Windows; imports checked
Windows test program without a console (`qt_add_executable` defaults to the GUI subsystem): results lost	`WIN32_EXECUTABLE FALSE`; QtTest log written to a file
Imported targets (OpenGL, DBus, Qt6Keychain) missing in `tests/keychain`	same `find_package` calls as `client/`
`QSKIP` called from a helper; “sign-out” phase without a stored session	macro in the test function; phase skipped unless storage is required
Android emulator: AVD not found	explicit AVD folder

Not tested

- Real Android phone (arm64) and a user’s Windows 10/11 computer (only the emulator and the CI’s Windows Server 2022).
- E-mail delivery through a real SMTP provider (local test relay only).
- Upgrading a 2.0 database holding real data to migration 0011 (tests apply 0001–0011 to an empty database); rollback to 2.0.
- 2.0 client against the 2.1 server; complete on-screen flows (build and static check only for the new dialogs).

Known limits of 2.1

- Windows Credential Manager keeps the session after uninstalling as long as the user has not signed out.
- The MinGW cross-build does not include QtKeychain: the session is never remembered.

- QtKeychain licence notice installed with the Windows applications, not yet shown in the APK.
- `log` transport: no code is delivered; `NH_EMAIL_TRANSPORT=smtp` is needed in production.
- Resetting a console account's password: out of scope (existing creators' procedure).

Validation 2.0.0

Date	27 October 2026
Version	2.0.0 (`project(nhill VERSION 2.0.0)`)
Repository	`Waynekaila/Nhill` (GitHub, private), branches `main` and `release/2.0.0`
Deliverables commit	`1c0f30c2fd33d6ac37b852f1f68adb9b154b2e97`
Principle	this report only mentions what was executed; everything else is under "Not tested" (§ 5)

1. Where checks took place

Environment	Role	Limits
Session workspace (Linux, without Qt or libpqxx development packages)	code changes, standalone `common` tests, static QML check, website, form backend (SQLite), local PostgreSQL 16 schema, documentation, PDFs	the server and Qt applications cannot be built there
GitHub Actions (`ubuntu-24.04`, `ubuntu:24.04` and `debian:trixie` containers, `windows-2022`)	full build, tests, packages, test installation	CI virtual machines, not a real user computer

Each job's results are published in `ci-results/*` branches of the repository (`RUN-INFO.txt`: workflow, status, commit, run link).

2. Tests executed and results

Suite	Where	Result	Evidence
`common` (amounts, taxes, codes, passwords, 128-bit integer)	CI Ubuntu 24.04 and Debian 13; workspace (standalone tool)	**22/22** passed; also 22/22 in `NHILL_PORTABLE_INT128` mode (Windows path) in the workspace	run 40464011312, `Testing/Temporary/LastTest.log`
`server` (API + throwaway PostgreSQL)	CI Ubuntu 24.04 (PostgreSQL 16), Debian 13 (PostgreSQL 17)	**83 passed, 1 skipped** (`Samples.GenerateDocumentSamples`, sample generation off by default) out of 84	same
`setup` (setup mode)	CI	**6/6**	same
`client` (Qt Test)	CI	**13/13**	same
Static QML check	workspace + CI	95 files, 0 problems, 0 untranslated texts	`scripts/qa/qml-check.py`
Website backend (SQLite)	workspace + CI	**13/13**	`website/backend/tests`
Website backend (PostgreSQL 16, real roles)	CI	**passed** (13 tests + read refused for the application role)	run 40461610891
Website pages	workspace	24 pages, 0 problems (links, `alt`, titles, FR/EN pairs, secrets)	`website/tools/check_site.py`
KPI workbook	workspace	91 formulas, 0 errors; values checked with test data	LibreOffice recalculation
Website server (smoke)	workspace	`/api/health` 200, form submission 201	session log

3. Deliverables built

Deliverable	Build	Checks	SHA-256
`linux/ubuntu-24.04/nhill-business_2.0.0-1_amd64.deb`	CI, `ubuntu:24.04` container	lintian 0 errors; `apt` install in a clean container, headless start, removal	`d46cadeb0701b6ff823ca65e8533116c6b1a372cd6a8bbbe078536cfa06d1c89`
`linux/ubuntu-24.04/nhill-console_2.0.0-1_amd64.deb`	same	same	`f26aa9f414343b8a80156b44f9891cce3e58614a060783caa4f9ccb438103e63`
`linux/ubuntu-24.04/nhill-pos_2.0.0-1_amd64.deb`	same	same	`c67f88b6af4c99bcf1ba3a8a3e6921e21c17e8ced11e41f0d6dbe2d2e1c34d3d`
`linux/ubuntu-24.04/nhill-server_2.0.0-1_amd64.deb`	same	lintian 0 errors, 7 warnings; installed in the clean container	`997c29c8c44cfe444ddc02bb21f2e2a0bcb6224339e76ab0222ecd2f8cba77ae`
`linux/debian-13/nhill-business_2.0.0-1_amd64.deb`	CI, `debian:trixie` container (Debian's Qt 6.8)	install, start, removal in a clean container	`63ce71aad7a5b67208384aaf1e740d4915561d978659183732e4ab949fe1d1d4`
`linux/debian-13/nhill-console_2.0.0-1_amd64.deb`	same	same	`579f363770d21e3c5660e920551d8f3864e03b084e3aea5f2ce8433a37f2bf87`
`linux/debian-13/nhill-pos_2.0.0-1_amd64.deb`	same	same	`ce804cd61b037b515a50d83ff1f513ca204798dea876b9ff58fc35d2a69f55b9`
`linux/debian-13/nhill-server_2.0.0-1_amd64.deb`	same	installed	`dc05552506a6b7cfcfe787d6f7abc9ffd8fb078e0efa4500b58939f154cc13d8`
`windows/nhill-business-2.0.0-setup.exe`	CI `windows-2022`, MSVC 2022, Qt 6.5.3, NSIS	PE format checked	`5a224a1b21f439cf6a63aba074830de8a3df56d6845e1ac80604ada7aeac2a1b`
`windows/nhill-console-2.0.0-setup.exe`	same	PE format checked	`e1f1c768b593a4097598e22e1c1a0e523b82c09de9c30b536638318b6b7e81b9`
`windows/nhill-pos-2.0.0-setup.exe`	same	PE format; **silent install and uninstall succeeded** on the runner	`b95c1dd8fae02f48157606637b1b185486abf8d04e9a5153bb4039027c5cdd07`
`android/nhill-business-2.0.0-arm64-v8a.apk`	CI, Qt 6.5.3 for Android, NDK r25c, platform 34	content (manifest, arm64-v8a libraries), **debug** signature verified by `apksigner`	`70c9664ad39baeadb61885d8d2ead5a6de19bb33d13d856f50d475e75f2fdf01`
Website (sources + archive)	workspace	checks of § 2	`release/V2.0.0/checksums/SHA256SUMS`
Documentation, legal and marketing PDFs	workspace (pandoc + Chromium)	`scripts/build-docs-pdf.sh --check`	`docs/pdf/manifest.json`

4. Issues found and fixed during validation

Issue	Fix
Windows build impossible: `__int128` unknown to MSVC in money computations	portable 128-bit integer (`common/include/nhill/int128.hpp`), compared with `__int128` on 20,000 operations; `ExchangeRate::scaled` as `int64_t`
`.deb` packages: build folder inside the source refused; logs counted in the source digest	build outside the source; `build/`, `dist/`, `release/` excluded from the digest
Debian 13: SVG plugin in a separate package, blocking package lookup	`qt6-svg-plugins` installed and declared, tolerant lookup
Windows: `Get-FileHash` unavailable in PowerShell 5.1 launched from pwsh; garbled accents	digests through .NET; script saved as UTF-8 with BOM
Android: CMake install rule invalid for the target; runner's platform 36 unreadable by Gradle 7	rule limited to desktop; platforms > 34 removed in CI
Console: no password check before sending at first access	check with the shared rule (8 characters)

5. Not tested (to be done by people, on real hardware)

Item	Why	Action
Windows installers on a user's Windows 10/11 computer (SmartScreen, antivirus, 150 % HiDPI, receipt printing)	CI runner only	manual acceptance (`docs/en/installation-guide.md` § 3.2)
Nhill Business and Nhill Console installed through their Windows installers	only Nhill POS is installed in CI	manual acceptance
APK on Android 7 and 14 phones/tablets	no device	manual acceptance; then release signing
Remembered sessions on Windows and Android	QtKeychain not included in the CI Windows/Android builds: sessions are not remembered there	add QtKeychain (vcpkg / Android build) then test
Production deployment with public TLS, production Supabase, backups, restore	no production infrastructure in the session	`docs/en/installation-guide.md` § 4
Real payments (Stripe, PayPal)	adapters tested against simulated responses only	test payment with a provider account
Nhill POS 2.0 screenshots	the automatic tour now reaches the choice of the demo till (created by `seed-demo.py`) but does not open the session: 2.0 sales screens are not captured automatically	adapt the tour (`PosMain.qml`) or capture by hand; Business 2.0 screenshots in `docs/assets/v2/`
Accessibility with a screen reader (NVDA, Orca, TalkBack)	not available	manual test
Legal	no validation by a lawyer	`legal/en/00-legal-research.md` § 6
External audit, penetration test	not ordered	before a wide opening

6. Known limitations of the release

- No forgotten-password reset (no e-mail sending).
- cpp-httplib 0.14 (dynamically linked to the distribution package): CVE-2025-46728 and CVE-2025-53629; mitigated by Caddy; upgrade to ≥ 0.23.0 recommended.
- Activity log: some detail values are shown raw (e.g. "Montant : 95000.00", "type : invoice") — seen in the 2.0 screenshots.
- Nhill is not certified for the DGI standardised invoice.
- The 8-character requirement is below the NIST recommendation (15) for a password without a second factor: accepted risk, to be recorded (`docs/en/security.md` § 3).

7. Human actions required before commercial launch

1. Create the operating company; complete every `[…]` field (`docs/en/legal-parameters.md`).
2. Have the 19 legal templates validated by a lawyer; check the competent data-protection authority (ARPTC / ARPTIC); formalities and transfers.
3. Acquire a Windows code-signing certificate and create the Android release key (`ANDROID_KEYSTORE_*` secrets); sign, then republish the digests.
4. Manual acceptance on real Windows and Linux computers and Android devices.
5. Choose production hosting; deploy server, Caddy, backups; test a restore.
6. Set prices after the pilot programme; enter them in Nhill Console.
7. Commission an external penetration test before a wide roll-out.
8. Check the availability of the “Nhill” trademark and register it.